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103
GEN

Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Engineering Chemistry

Subject Code : BSC 103

Time Allowed : 3 Hours

Full Marks : 70

Pass Marks : 21

Answer in your own words.
Answer any five *questions in which* Question No. 1 *is compulsory.*
The figures in the margin indicate full marks.
All questions carry equal marks.

1. Choose the correct answer of the following questions: 2×7=14

(i) The fundamental particle with no charge is
(a) Electron
(b) Proton
(c) Neutron
(d) Positron

(ii) The rule used to fill electrons in increasing order of energy is
(a) Pauli Principle
(b) Hund's Rule
(c) Aufbau Principle
(d) Bohr Rule

(iii) The electrolyte that completely dissociates in aqueous solution is
(a) Weak electrolyte
(b) Strong electrolyte
(c) Non-electrolyte
(d) Organic acid

(iv) The process used to extract metal from ore by reduction is
(a) Concentration
(b) Refining
(c) Smelting
(d) Roasting

(v) Bakelite is prepared by
(a) Addition polymerization
(b) Condensation polymerization
(c) Copolymerization
(d) Step-growth reaction

(vi) The material used for thermal insulation is
(a) Steel
(b) Glass wool
(c) Copper
(d) Brass

(vii) The major pollutant responsible for greenhouse effect is
(a) SO
(b) CO
(c) NH
(d) None of these

Please Turn Over

2. (a) Define atom. Explain Bohr's model and limitations.

   (b) Explain formation of covalent and ionic compounds with examples. 7+7

3. (a) Define ionisation. Explain Arrhenius theory of ionisation.

   (b) Explain primary and secondary reactions during electrolysis with examples. 7+7

4. (a) What are metals? Explain mechanical properties of metals.

   (b) Describe the extraction process of any two metals. 7+7

5. (a) Explain preparation, properties and uses of alloys.

   (b) Write notes on Duralumin, German Silver and Monel Metal. 7+7

6. (a) Explain polymerization. Distinguish between thermosetting and thermoplastic polymers.

   (b) Explain natural rubber, its vulcanisation and applications. 7+7

7. Write short notes on *any four* of the following: 3·5×4=14

   (a) Electrochemical Series

   (b) Electrolysis of NaCl

   (c) Synthetic Polymers

   (d) Water Pollution — causes and control

   (e) Thermal Insulating Materials
